

Question	Answer	Mark	Mark scheme	Additional guidance
24 (a)	1 : 4	P1	for process to equate the two volumes, $\frac{4}{3}\pi r^3 = \frac{1}{3}\pi r^2 h$	Can be implied by $3r = l$
		A1	cao	
(b)	$1 : \sqrt{8}$	P1	for process to equate surface areas, eg $4\pi r^2 = \pi r^2 + \pi r l$	
		P1	for process to substitute $l = \sqrt{h^2 + r^2}$, eg $4\pi r^2 = \pi r^2 + \pi r \sqrt{h^2 + r^2}$	
		P1	for process to isolate term in r^2 after substituting for l , eg $8r^2 = h^2$	
		A1	for $1 : \sqrt{8}$	